Econometrics Midterm · Short Answer

1. What is the difference between a population and a sample?

2. Why do econometricians usually study a sample instead of the population, and what is the goal of studying the sample?

3. Define external validity and state which property of the sample matters for it.

4. In log(price) = β_0 + β_1 subsidy + β_2 log(sunlight) + u, why is β_1 called a partial slope rather than simply "the slope"?

5. In log(price) = β_0 + β_1 battery + β_2 sunlight + β_3 battery×sunlight + u with $\beta_3 \neq 0$, why does battery not have a linear relationship with log(price)?

6. What is the difference between random sampling and random assignment?

7. For Model 2, wage = β_0 + β_1 educ + u (ability not included), explain why the zero conditional mean condition may be violated.

8. What is a control variable, and why is ability added in Model 1, wage = β_0 + β_1 educ + β_2 abil + u?

9. What does "ceteris paribus" mean, and why is it important in econometrics?

10. What is panel (longitudinal) data, and what is one advantage of it?

11. What does OLS minimize, and what is a residual?

12. What does R-squared measure in a regression?

13. What does the homoskedasticity assumption (SLR.5) state?

14. What does it mean that the OLS estimator $\hat{\beta}_1$ is unbiased?

15. When does omitting a relevant-looking variable x_2 NOT cause bias in the estimated coefficient on x_1?

16. What is the main consequence of multicollinearity (high but not perfect correlation among regressors) for OLS?

17. State the Gauss–Markov theorem.

18. Define the p-value of a hypothesis test.

19. Does a statistically significant coefficient always matter in practice? Answer in 1–2 sentences.

20. Interpret a 95% confidence interval for β_j.

21. Which can make the usual OLS t statistics invalid (no t distribution under H_0): (i) heteroskedasticity, (ii) a sample correlation of 0.95 between two regressors in the model, (iii) omitting an important explanatory variable? Explain briefly.

22. What does it mean that an estimator is consistent, and which assumptions are enough for OLS to be consistent?

23. pctstck = $\beta_0 + \beta_1$ funds + β_2 risktol + u satisfies MLR.1–MLR.4 (risktol = risk tolerance). If funds and risktol are positively correlated and pctstck is regressed on funds only, what is the sign of the inconsistency in $\tilde{\beta}_1$, and why?

24. Can cigs (average cigarettes smoked per day) have a normal distribution in the adult population? What does this imply for OLS inference?

25. How is the LM (n-R-squared) statistic for testing q exclusion restrictions computed, and what distribution does it follow?

26. If the dependent variable is divided by 1,000 (dollars → thousands of dollars), what happens to the OLS coefficients, standard errors, t statistics and R-squared?

27. What is a beta (standardized) coefficient, and why is it useful?

28. For rdintens = 2.613 + 0.00030 sales − 0.0000000070 sales^2 (sales in $ millions), give the turning-point formula for a quadratic and find where the marginal effect of sales becomes negative.

29. In log(wage) = $\beta_0 + \beta_1$ educ + β_2 educ· pareduc + β_3 exper + β_4 tenure + u (pareduc = both parents' education), what is the return to one more year of education, and what does β_2 measure?

30. Why is adjusted R-squared used, and when does adding a variable increase it?

31. Why is a 95% prediction interval for one individual outcome y^0 much wider than the 95% confidence interval for $E(y \mid x = x^0)$?

32. When the dependent variable is log(y), why is exp(predicted log(y)) not a good prediction of y, and how is it fixed?

Answers

1. A population is the **entire** set of individuals, objects or data points we want to study, while a sample is a **subset** of the population that is **selected** for analysis.
Keywords: entire · subset · selected

2. Studying the whole population is often too **costly** or **impossible**, so we study a sample; the goal is to find out what the **population is like** from the sample.
Keywords: costly · impossible · population is like

3. External validity is the validity of applying the **conclusions** of a study **outside** the context of that study; it depends on whether the sample is **representative** of the population.
Keywords: outside · conclusions · representative

4. β_1 is the change in log(price) from a **one-unit** increase in subsidy while **holding** the **other variables** (log(sunlight)) fixed, so it isolates the effect of subsidy alone.
Keywords: holding · other variables · one-unit

5. Because battery and sunlight **interact**, the effect of battery on log(price) is $\beta_1 + \beta_3 \cdot$sunlight, which **depends** on the level of sunlight; the slope is **not constant**.
Keywords: interact · depends · not constant

6. In random sampling every element of the population has an **equal probability** of being **selected**; in random assignment every element has an equal probability of being **assigned** to a group.
Keywords: equal probability · selected · assigned

7. Ability is omitted, so it goes into the **error** u; because people with better ability get more education, educ is **correlated** with u, so $E(u|educ) = 0$ fails and β_1 is estimated with bias.
Keywords: ability · error · correlated

8. A control variable is included to **hold** another factor **fixed** so it no longer sits in the error; adding ability helps satisfy the **zero conditional mean** condition and removes the **bias** in β_1.
Keywords: hold · zero conditional mean · bias

9. Ceteris paribus means "**other relevant factors** being **equal**"; most economic questions ask for the **causal** effect of one variable holding all other factors fixed.
Keywords: other factors · equal · causal

10. Panel data follow the **same** cross-sectional units **over time**; they can account for **time-invariant** unobservables and model lagged responses.
Keywords: same · over time · time-invariant

11. OLS chooses the estimates that minimize the **sum of squared residuals**; a residual is the **actual** value y_i minus the **fitted** value $\hat{y}_i$.
Keywords: sum of squared residuals · actual · fitted

12. $R^2 = SSE/SST$ is the **fraction** of the sample **variation** in y that is **explained** by the explanatory variable(s).
Keywords: fraction · variation · explained

13. Homoskedasticity means the **error** u has the **same** (constant) **variance** for any value of x: $Var(u|x) = \sigma^2$.
Keywords: variance · error · same

14. Unbiasedness means $E(\beta_1) = \beta_1$: if sampling and estimation were **repeated** many times, the estimates would **on average** equal the **true** population value, though one estimate may differ.
Keywords: on average · repeated · true

15. There is no omitted variable bias if the omitted variable is **irrelevant** (its coefficient is zero) or **uncorrelated** with x_1.
Keywords: irrelevant · uncorrelated

16. Multicollinearity **increases** the sampling **variance** of the affected slope estimators, but OLS remains **unbiased**; it can be detected with variance inflation factors.
Keywords: variance · increase · unbiased

17. Under **MLR.1–MLR.5**, OLS is **BLUE** (best linear unbiased estimator): it has the **smallest variance** among all linear unbiased estimators.
Keywords: BLUE · smallest variance · MLR.1

18. The p-value is the **smallest significance level** at which the null hypothesis would be **rejected**; we reject H_0 when the p-value is below the chosen level.
Keywords: smallest · significance level · reject

19. Not necessarily: statistical significance only shows the effect differs from zero, while **economic** (practical) significance depends on the **magnitude** of the coefficient.
Keywords: not necessarily · magnitude · economic

20. In **repeated** samples, intervals built this way **contain** the **true** population coefficient in 95% of cases; the bounds are random, not the parameter.
Keywords: repeated · contain · true

21. Heteroskedasticity (i) and an **omitted** important variable (iii) violate the CLM assumptions, so the t statistics are invalid; a high **correlation** of 0.95 (ii) is not a CLM violation (only perfect collinearity is ruled out).
Keywords: heteroskedasticity · omitted · correlation

22. An estimator is consistent if, as the **sample size** grows, its distribution **collapses** to the **true** parameter (plim $\tilde{\beta}_j = \beta_j$); OLS is consistent under **MLR.1–MLR.4** (even zero correlation between each x_j and u is enough).
Keywords: sample size · collapse · true · MLR.4

23. The inconsistency is **positive**: plim $\tilde{\beta}_1 - \beta_1 = \beta_2\delta_1$, with $\beta_2 > 0$ (higher **risktol** raises pctstck) and $\delta_1 > 0$ because funds and risktol are positively **correlated**; it does **not disappear** as n grows.
Keywords: positive · risktol · correlated · not disappear

24. No: most adults smoke **zero** cigarettes and cigs is **nonnegative** and discrete, so it is **not normal**; with a **large sample**, asymptotic normality still makes the t and F tests approximately valid.
Keywords: zero · nonnegative · not normal · large sample

25. Estimate the **restricted** model, regress its **residuals** on all the explanatory variables, and compute LM = $\mathbf{nR^2}$ from that auxiliary regression; under H_0 it is approximately **chi-square** with q degrees of freedom.
Keywords: restricted · residuals · nR · chi-square

26. All coefficients and standard errors are **divided by 1,000**, while the **t statistics** and **R-squared** stay **unchanged**; only the units change, not the statistical conclusions.
Keywords: divided · unchanged · t statistic · R-squared

27. A beta coefficient comes from regressing the **standardized** y on standardized x's; it is the change in y, in **standard deviations**, from a one standard deviation increase in x_j, so effects measured in different **units** can be **compared**.
Keywords: standardized · standard deviation · units · compare

28. Turning point x^* = $\mathbf{\hat{\beta}_1/(2|\hat{\beta}_2|)}$ = 0.00030/(2 × 0.0000000070) ≈ **21,428.57**, i.e. sales of about **$21.4 billion**; beyond that the estimated marginal effect of sales on rdintens is negative.

29. The return is $\beta_1 + \beta_2 \cdot$pareduc, so it **depends** on **parents' education**; the **interaction** coefficient β_2 is how much the return changes for one more year of pareduc (estimate 0.00078: at pareduc = 32 the return is about 7.2%).
Keywords: depends · parents · interaction

30. R-squared **never falls** when a variable is added, so adjusted R-squared imposes a **penalty** for extra regressors using **degrees of freedom**; adding a variable raises it only if that variable's **t statistic** exceeds one in absolute value.
Keywords: never falls · penalty · degrees of freedom · t statistic

31. The confidence interval reflects only the sampling **variance** of $\hat{y}^0$, while the prediction interval also includes the variance of the unobserved **error** for an **individual** outcome (σ^2), which does not shrink as n grows (e.g. colgpa: CI 2.66–2.74 vs PI 1.60–3.80).
Keywords: error · variance · individual

32. exp(log $\hat{y}$) systematically **underestimates** E(y|x); we **multiply** it by $\hat{\alpha}_0 = \exp(\hat{\sigma}^2/2)$ under **normal** errors, or by the average of $\exp(\hat{u}_i)$ from the residuals.
Keywords: underestimate · multiply · residuals